



Project Based Learning Report on AgriWatch: Intelligent Crop Monitoring System for the course IoT and Applications (IS244AI)

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in
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DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING



CERTIFICATE

Certified that the project work titled **“AgriWatch: Intelligent Crop Monitoring System”** is carried out by **Shreyas Sahoo (1RV24IS122)** and **Samarjeet Bhonsle (1RV24IS107)** in partial fulfilment of the completion of the course **IoT and Applications (IS244AI)** of the IV Semester, B.E., in **Information Science and Engineering**, during the academic year 2025-26. It is certified that all corrections/suggestions indicated for the Laboratory Internal Assessment have been incorporated in the project report and duly approved by the faculty. The project report has been approved as it satisfies the academic requirements in respect of internship project work prescribed by the institution for the said degree.

Signature of Faculty In-charge

Signature of HoD

External Viva

Name of Examiner

Signature with Date

Abstract of AgriWatch: Intelligent Crop Monitoring System

AgriWatch: Intelligent Crop Monitoring System is an IoT-based smart agriculture solution designed to enhance farming efficiency through real-time environmental monitoring and remote field inspection. The system consists of two major components: a static monitoring unit and a dynamic rover unit. The static unit employs an ESP32 microcontroller integrated with soil moisture, pH, and DHT11 temperature-humidity sensors to continuously monitor critical agricultural parameters. The collected data is transmitted wirelessly to a centralized dashboard, enabling farmers to observe field conditions remotely.

The dynamic unit consists of a rover equipped with an ESP32 controller, ESP32-CAM module, motor driver, and DC motors. The rover can be remotely controlled through the dashboard and provides live video streaming for visual crop inspection. This allows farmers to monitor crop health, detect abnormalities, and inspect inaccessible areas without physically entering the field.

The dashboard serves as a unified interface for displaying sensor readings, controlling rover movement, and viewing live camera feeds. By combining IoT sensing, wireless communication, and mobile surveillance, the proposed system supports precision agriculture practices, improves resource utilization, reduces manual effort, and enables timely decision-making. The solution is cost-effective, scalable, and suitable for both small-scale and large-scale farming applications.

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1. Introduction

The IoT-Based Smart Agriculture Monitoring and Rover System is an innovative farming solution that helps increase productivity through real-time informational data and remote inspections of various types of farmland. The IoT-Based Smart Agriculture Monitoring and Rover System contains IoT technology, wireless communication, sensor networks, and mobile rovers enabling farmers with up-to-date information to help them make compliance or non-compliance decisions relating to their crops and/or the conditions of their soil.

The IoT-Based Smart Agriculture Monitoring and Rover System has two main parts: a stationary monitoring unit and a mobile rover unit. The stationary unit uses an ESP32 microcontroller, is fixed within the farmland at an established location, and continually collects important environmental data using sensors, such as a soil moisture sensor, pH sensor, and DHT11 temperature and humidity sensor. These parameters are critical in determining soil fertility, water requirements and help in determining overall crop growth conditions. Once the information is collected, it is wirelessly transmitted to an on-line dashboard which allows a farmer to monitor the farmland conditions from a remote location and in real-time.

A rover is a moving platform or vehicle that consists of a microcontroller (ESP32) a camera module (ESP-CAM), and a motor driver. A web dashboard utilizing the directional controls of the dashboard can be used to control the rover from a distance to navigate different areas of a farm. The live video feed provided by the ESP-CAM allows for visual inspections of crops and field conditions without physically being present in the field. This capability will allow you to identify plant diseases, pest problems, irrigation issues, and other defects in your crops.

This project will create a smart farm application using an integrated system that combines sensor-based monitoring with a mobile surveillance platform to reduce labor requirements, enhance resource management efficiency, and provide support for precision agriculture. The G3610 will provide real-time data acquisition capabilities, remote access to the field, mobility in the field, and a low-cost and scalable solution for modern agriculture (both large-scale and small-scale) that will improve productivity and sustainability.

2. Existing System Study and Project Objectives

2.1 Existing System Study

Agriculture has undergone significant technological advancements with the introduction of Internet of Things (IoT) technologies. Several existing smart farming systems utilize environmental sensors to monitor soil moisture, temperature, humidity, and other parameters to assist farmers in managing agricultural activities. These systems provide real-time monitoring capabilities and help improve crop productivity through data-driven decision making.

Many modern agricultural monitoring solutions employ wireless sensor networks and cloud-based dashboards for remote access to field data. While these systems effectively collect environmental information, most of them are limited to fixed monitoring stations and lack mobility for detailed field inspection.

In recent years, robotic systems and agricultural rovers have also been introduced for crop monitoring applications. These rovers provide mobility and visual inspection capabilities through onboard cameras. However, many existing rover-based systems primarily focus on surveillance and do not integrate environmental monitoring sensors into a unified platform.

The survey of existing solutions reveals that most systems focus either on environmental monitoring or on visual inspection. Very few systems provide both capabilities together in a cost-effective and scalable architecture suitable for small and medium-scale farming.

2.2 Outcome of existing system study

The literature survey and analysis of existing agricultural monitoring systems highlighted several limitations:

- Lack of integration between environmental monitoring and visual crop inspection.
- Dependence on manual field visits for detailed crop observation.
- Limited mobility in fixed sensor-based monitoring systems.
- High implementation costs for advanced agricultural monitoring solutions.
- Difficulty in continuously monitoring large agricultural fields.

These limitations motivated the development of an integrated system capable of combining real-time environmental monitoring with remote field inspection.

2.3 Project Objectives

The primary objectives of the proposed AgriWatch system are:

- To continuously monitor soil moisture, pH, temperature, and humidity conditions in agricultural fields.
- To provide real-time access to environmental data through a web-based dashboard.
- To develop a mobile rover capable of navigating agricultural fields remotely.
- To provide live video streaming for crop inspection and field surveillance.
- To reduce manual effort involved in field monitoring.
- To improve decision-making through real-time agricultural data.
- To create a cost-effective and scalable solution suitable for modern precision agriculture.

3. Project Architecture and Design

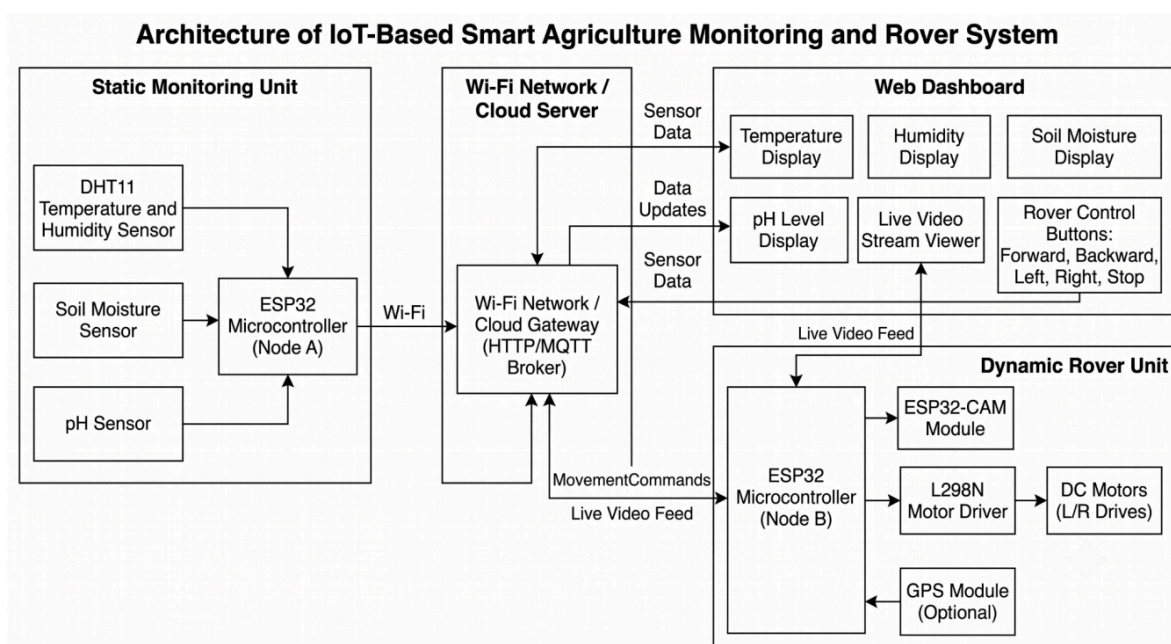
The overall system consists of two major systems: Static Monitoring Unit and Dynamic Rover Unit. Both units are able to communicate with a central dashboard wirelessly, so the user can monitor the conditions in the field and remotely control the rover.

The Static Monitoring Unit consists of an ESP32 microcontroller connected to a soil moisture sensor, pH sensor and DHT11 temperature-humidity sensor. The sensors continuously take environment and soil parameters from the farmland, and the ESP32 processes the data and sends it to the monitoring dashboard for visualization and analysis.

The Dynamic Rover Unit consists of an ESP32 microcontroller, ESP32-CAM module, motor driver and DC motors. After receiving movement commands from the dashboard, the rover will navigate through the agricultural field. Also, the ESP32-CAM will provide live video streaming for remote crop inspection and surveillance. The overall data flow in the entire system is generally summarised as follows:

- Environmental and soil data is acquired in near real-time via sensors.
- Acquired sensor reading values are processed and transmitted via the ESP32.
- The user is able to view the acquired sensor values via the dashboard for monitoring purposes.
- The user is able to give the rover movement commands via the controls on the dashboard.
- The rover receives the commands and moves as directed.
- The ESP32-CAM is providing a live video stream via the dashboard of the field.
- The user can use both sensor data and visual inputs to make better farming decisions.

System Architecture Diagram:



3.1 Static Monitoring Unit

The monitoring unit for the environment and soil parameter data is known as the static monitoring unit. It collects data at a specified set point in the field, continually collecting variables that determine productivity from agriculture.

Soil moisture sensors measure the amount of water in the soil and can indicate crop irrigation. The pH sensor indicates the degree of acidity or alkalinity found in the soil and is able to help provide fertility and nutrient availability for the soil. The DHT11 sensor measures both the temperature and relative humidity of the environment and directly affects how well plants will grow in an area.

An ESP32 microcontroller is placed and soil conditions will be collected by the sensors connected to the system, and then processed via an ESP32 to send the readings to a dashboard. From there, users will be able to access the readings remotely via the dashboard to verify field conditions and crop health.

In the event that a visual inspection needs to take place, the user may issue a movement command through the dashboard to tell the rover where to go. Once the rover has reached its intended location, it will stream real-time video back to the user. The user can then use the sensor data and the visual observations from the rover to make informed decisions about irrigation, fertilization, or other crop maintenance activities.

This integrated process for environmental monitoring and surveillance of the fields is part of a smart agriculture solution, which will allow farmers to improve productivity and use their resources more efficiently.

Actions as the central processing unit or processor of the monitoring unit, taking the sensor measurement data and converting them into digital values. The ESP32 then communicates with the display and provides real-time and continual information related to the environmental conditions in the field

3.2 Dynamic Rover Unit

Rover Unit (Dynamic) for remote mobility and visual inspections. Rover Unit differs from Static Monitoring Unit because it has the capability of moving to different areas on the farm and allowing users to visually inspect crops in multiple areas. The rover uses an ESP32 Micro-Controller, which is connected to a motor driver and DC motors to control the movement based on directional commands (forward, backward, left, right, stop) sent from the dashboard via a wireless network. Once the ESP32 receives the command, it uses the received command to control the motors on the rover for the corresponding motion.

To provide a real-time video stream, the rover is equipped with an ESP32-CAM module. This allows the farmers to use the rover to observe the crop's condition, identify irrigation problems, find evidence of pests, and check for plant health without physically going into the field.

3.3 Dashboard and Communication Module

The dashboard acts as the hub or central point of interaction between the user and the ag monitoring system. It is where sensor data is displayed as it is collected in real-time and includes the buttons required for using the rover to help with navigation.

The communication that takes place between the different parts of the system occurs via Wi-Fi. The ag monitoring unit sends sensor data to the dashboard. The dashboard sends the user-generated commands back to the rover. This bi-directional communication creates a means to enable seamless and effortless interaction between monitoring and controlling parts of the system.

The dashboard provides a single point of access to monitor the environment and view live video feeds from the fields so that farming operations can be accomplished easily and quickly.

3.4 System Workflow

Environmental and soil conditions will be collected by the sensors connected to the system, and then processed via an ESP32 to send the readings to a dashboard. From there, users will be able to access the readings remotely via the dashboard to verify field conditions and crop health.

In the event that a visual inspection needs to take place, the user may issue a movement command through the dashboard to tell the rover where to go. Once the rover has reached its intended location, it will stream real-time video back to the user. The user can then use the sensor data and the visual observations from the rover to make informed decisions about irrigation, fertilization, or other crop maintenance activities.

This integrated process for environmental monitoring and surveillance of the fields is part of a smart agriculture solution, which will allow farmers to improve productivity and use their resources more efficiently.

4. Methodology

This section discusses the practical realization of the suggested Smart Agriculture Monitoring and Plant Disease Detection System. The suggested system is an amalgamation of an IoT-enabled environmental monitoring, rover-assisted field inspection, and AI-based plant disease detection system on a singular platform. The implementation revolves around the factors related to the integration of hardware components, communication mechanism, image acquisition, and dashboard functions. The modular design of the system ensures that each module has its own specific task and, at the same time, fits together into the entire system.

4.1 Implementation Details

Two main components make up the proposed system: the Static Monitoring Unit (SMU) and the Dynamic Rover Unit (DRU). Both units communicate with a central dashboard via wireless communication.

The SMU is installed in a constant position in an agricultural field that takes continuous measurements of several environmental parameters and soil properties. The DRU is mobile and can move around the field to visually inspect crops and take photos of crop damage due to disease for diagnostics. The photos are processed through a disease detection model, and the results (along with all of the other SMU measurements) will be displayed on the central dashboard.

The dashboard serves as the primary control interface for the entire system and allows the user to easily view all sensor measurements, the status of the rover, video feed from the rover's camera feed, as well as all results from the disease prediction program.

4.2 Static Monitoring Unit

The static monitoring unit comprises of an ESP32 microcontroller connected to three sensors: a soil moisture sensor, a pH sensor and a DHT11 temperature and humidity sensor. The ESP32 will gather data from the sensors in real-time (every second), transmitting this data back to the dashboard via Wi-Fi.

The soil moisture sensor allows the farmer to assess their irrigation needs as it provides a measurement of the amount of moisture in the soil and helps them to estimate the quantity of water required for irrigation purposes. The pH sensor provides data regarding the pH level of the soil – whether it is acidic or alkaline – which affects how well nutrients are absorbed by plants and, therefore, is a critical factor in the growth of crops. The DHT11 temperature and humidity sensors record the temperature and relative humidity of a specific environment.

The information gathered from these three sensors will be recorded and updated in real time, displayed on the dashboard, and used to aid in making informed decisions for the management of crops and irrigation of fields. Farmers will have remote access to their fields through the dashboard, allowing them to monitor their fields and make decisions on the fly.

4.3 Dynamic Rover Unit

The Dynamic Rover Unit is a remote crop inspection tool that utilizes the ESP32 microcontroller, L298N motor driver module, and DC motors mounted on a chassis, among other components such as the ESP32-CAM module.

The user will have remote access to control the rover in all directions via the dashboard using directional controls. Based on the user's input, the ESP32 will produce control signals to drive the motors through the driver module.

This rover could allow for more efficient operation within an agricultural field and allow for crop inspection over much larger areas than could ordinarily be accomplished through manual field visits. This is especially helpful for large farms, where the need for continuous monitoring of crops would otherwise be very challenging.

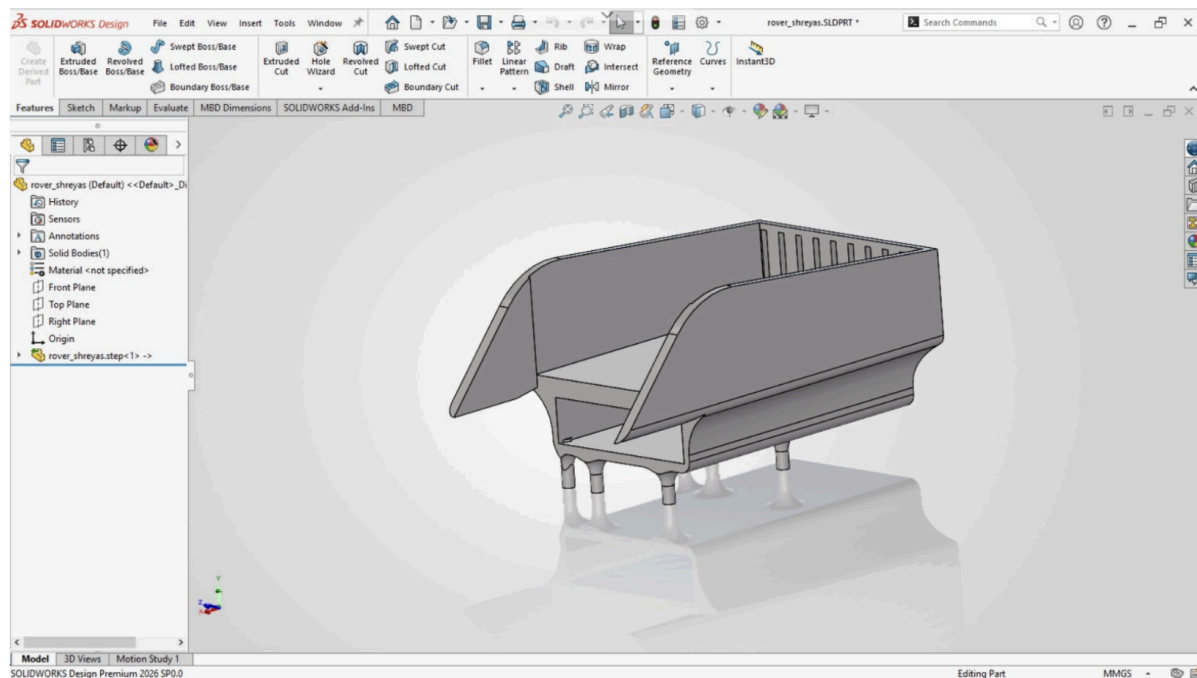


Fig 4.1 (a): CAD Design of Rover Part

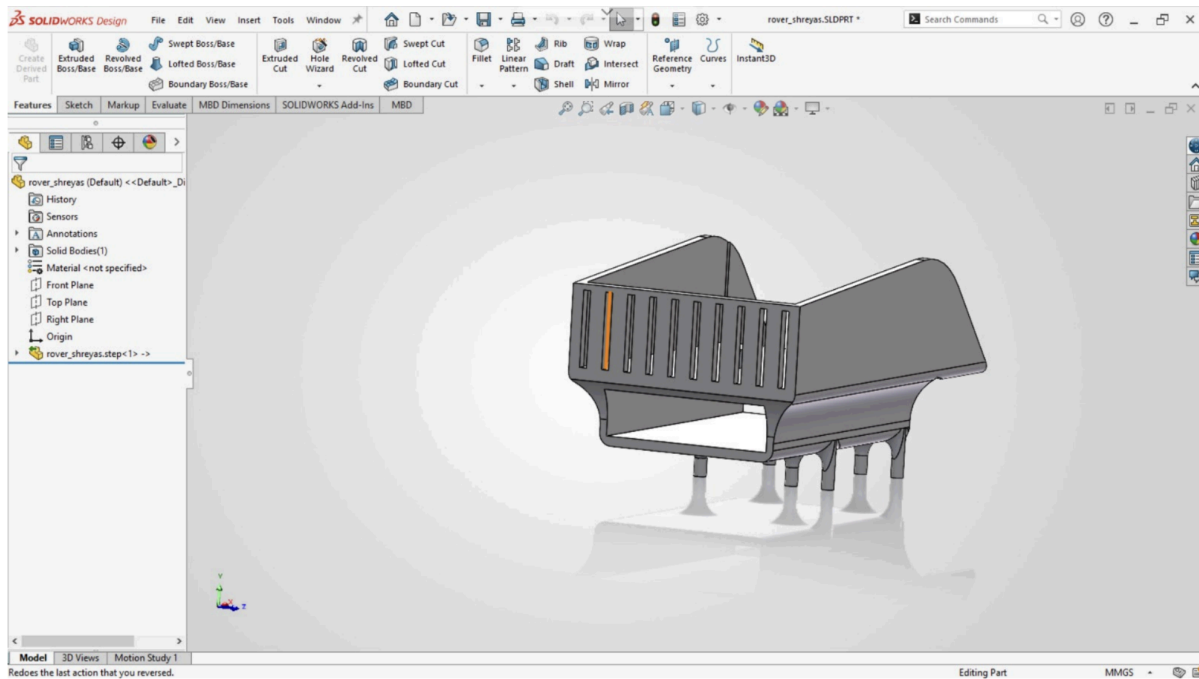


Fig. 4.1 (b): CAD Design of Rover Part

4.4 ESP-32 CAM Integration

The image acquisition component of the remainder of the system is created with the use of the ESP32-CAM module mounted on the rover.

By taking various images of crop leaves, the camera will provide real-time streaming to a dashboard for farmers to see crop conditions remotely and look for visible signs of disease, insects, or nutrient deficiency.

Images captured will be sent to a disease detection module for analysis. The live video feed will assist users with navigating the rover to locate areas that need further inspection.

4.5 Disease Detection Module

A Convolutional Neural Network (CNN) model is employed to identify plant disease (plant/pathogen identification) utilizing captured images of the leaf.

A dataset containing images of both healthy and diseased plants is provided to train the model. After training, the model processes images captured by the ESP32-CAM, classifying them into one of the many plant disease classifications using a confidence score.

After being classified by the trained CNN, the results are sent to a dashboard for display for the farmer, allowing them to identify and take action on the disease prior to the disease spreading to other crops.

4.6 Dashboard Implementation

A centralised interface is presented via the web-based dashboard for monitoring the system and controlling its operation.

The dashboard shows the current information (i.e. values) regarding soil moisture, soil pH, temperature and humidity in real time, displays the movement control for rover navigation, and displays the live camera feed from the ESP32-CAM.

Furthermore, the dashboard also displays the results of disease detection, providing the predicted disease name and crop health status. This integration provides users with access to all system functions via one interface.

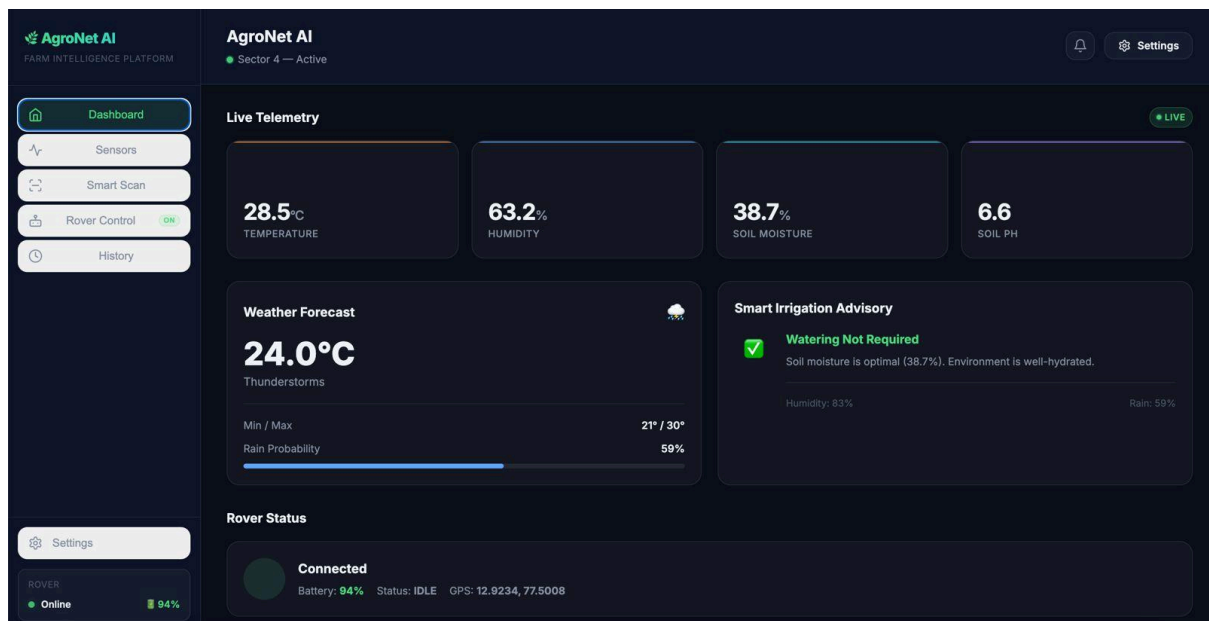


Fig. 4.2 (a): Web Dashboard

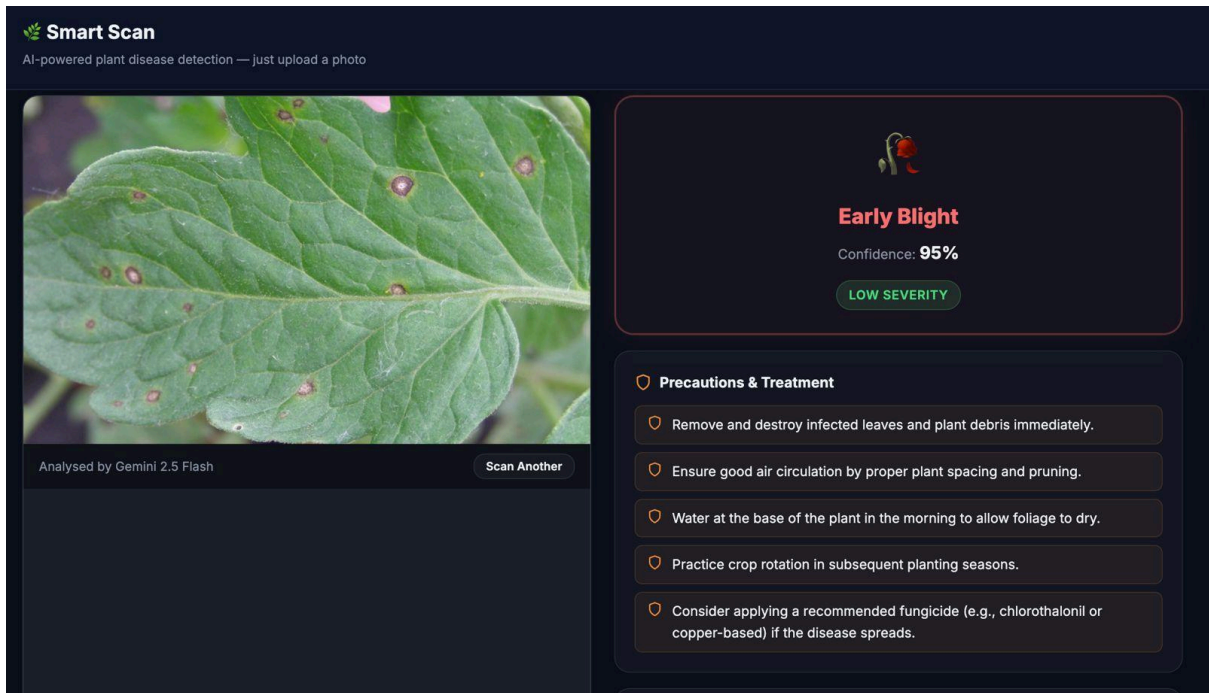


Fig. 4.2 (b): Scanning of plant for disease detection

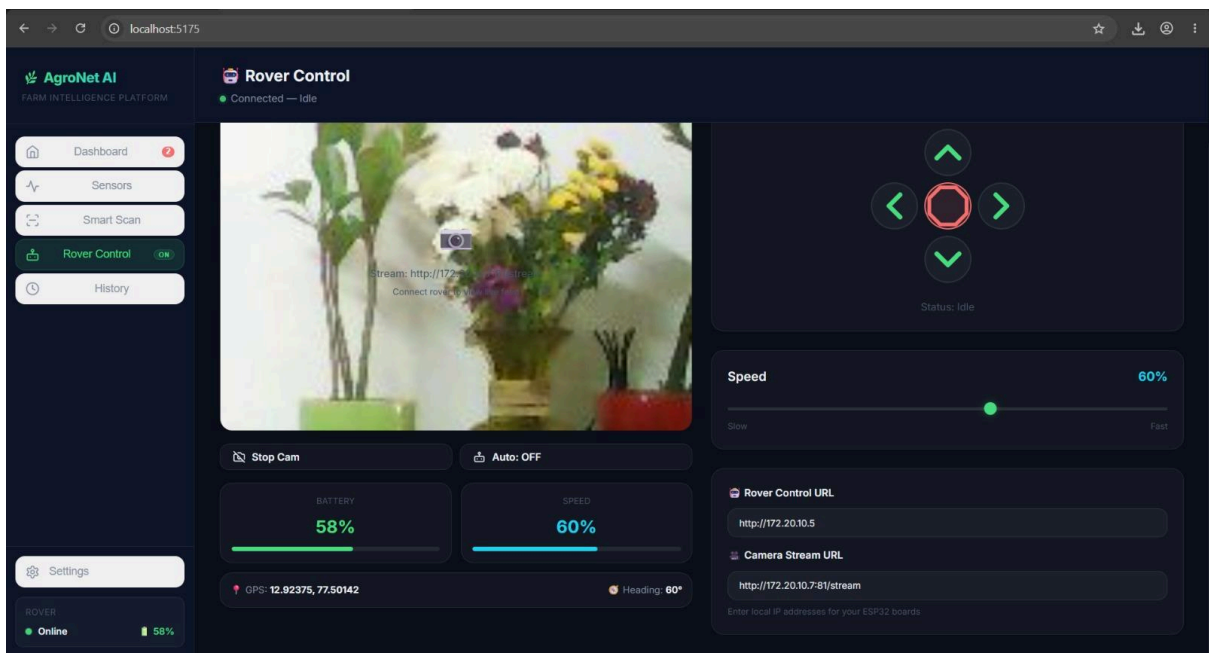


Fig. 4.2 (c): Live ESP32 camera feed

4.7 System Workflow

A system operates based on multiple connected processes in a series. There is an environmental and soil monitoring unit that collects data continuously (static monitoring unit) and sends that data to an interactive dashboard. The rover can then be operated through this same interactive dashboard to inspect crops using live camera feeds from the rover.

Images collected by the rover are sent to a disease detection module where the CNN model analyzes the images and predicts the occurrence of any possible plant diseases. The results of this prediction, as well as the sensor values and live images from the rover, are sent back to and displayed on the interactive dashboard.

This workflow allows for constant monitoring, efficient field inspections, and timely detection of diseases in a field.

4.8 Summary

Discussion in this chapter covered all of the key details related to how to implement the Smart Agriculture Monitoring and Plant Disease Detection System. The system combines many different types of IoT Sensors, ESP32-Controller based hardware, Rover mobility, Image acquisition, and AI-based Disease detection into one cohesive system. The Dashboard is the primary control & monitoring tool for observing field conditions in real-time and diagnosing diseases before they become a problem. The modular design makes sure that the overall system will function in an efficient manner, allowing the implementation of a scalable solution for today's precision agriculture.

5. Results of AgriWatch

This chapter presents the results obtained from the implementation and testing of the proposed IoT-based Smart Agriculture Monitoring and Disease Detection System. The performance of both the static monitoring unit and the rover-based dynamic unit was evaluated under different operating conditions. The static unit was responsible for continuously monitoring environmental and soil parameters, while the rover unit enabled remote navigation, image acquisition, and location tracking. The collected data was transmitted to the dashboard in real time, allowing users to monitor field conditions and access crop health information. The obtained results demonstrate the effectiveness of integrating sensor networks, wireless communication, image processing, and intelligent recommendation systems into a unified agricultural monitoring platform.

5.1 Execution Outcomes

A controlled environment was successfully deployed where by the entire system has been thoroughly tested. A stand-alone monitoring device throughout the system has continuously collected temperature, humidity, soil moisture, and soil pH data while also sending data to an online dashboard via a wireless network in real-time. Because every sensor's reading was updated multiple times daily, field conditions were continuously monitored without manual intervention. The dashboard received and displayed all environmental data, as well as soil-related data from the ESP32 sensor node. The live monitoring feature of the dashboard allowed the user to see how conditions (both climate and soil) change over time, resulting in highly informative data that can assist the user when making decisions regarding the management of irrigation and crops.

[4.1]

The rover-based subsystem was controlled via a web interface, whereby users can control all rover movement actions (forward, backward, left, right and stop) from a remote location. By implementing a control mechanism with very low latency rates and extending the usage of the control system with very stable communication during the testing phase, the user was able to perform control operations seamlessly.

[4.2]

The ESP32-CAM module captured images of the user's plants and sent them to the dashboard; once on the dashboard the image was processed through the disease classification module which identifies the type of disease providing the user with disease-specific recommendations for treatment and fertilizer. The Gemini API was instrumental in generating detailed treatment and fertilizer recommendations based on the specific condition of the plant with respect to the disease. The GPS-enabled rover also had the ability to identify its geographic coordinates by using the NEO-6M GPS module. The geographical coordinates collected by the GPS receiver were successfully sent to a designated point and displayed on the dashboard. Because of this function, observations regarding diseases and inspections performed in the field can be tied to specific geographic coordinates within the farmland.

[4.3]

The ability to operate both the static unit and dynamic unit together proved that it is possible to give comprehensive field monitoring from a distributed IoT design. The complete integration of real-time data collection, wireless connectivity, image analysis, and intelligent recommendations into one platform was achieved.

5.2 Handling Edge Cases

The proposed solution was subjected to a range of edge case scenarios in testing, which were designed to assess the proposed system's robustness and reliability. One of the edge cases examined the impact of the temporary availability of the wireless network. The testing facility showed that the modules (ESP32) would automatically reconnect to the wireless network as soon as the network became available, and this occurred without having to perform any manual resets. Another edge case examined was related to the performance of the environmental sensors, which may be impacted by external factors. Environmental sensor readings may fluctuate due to changes in environmental conditions or due to temporary interference with the sensor's operation. The system continuously updated and refreshed the sensor values so that the dashboard provided the most recent observations.

The third edge case evaluated was the GPS subsystem's ability to determine valid coordinates under disparate conditions, such as when satellite signals are not available. Upon initial startup, the GPS subsystem did not provide a valid coordinate and was not able to provide a valid coordinate while operating in an area with limited visibility of the sky or with temporary loss of satellite signals.

When the system acquired satellite signals, it was able to quickly determine and report valid coordinates in real-time. The image acquisition system was evaluated under many different conditions, including analysing images from the ESP32-CAM in full daylight with sufficient clarity to identify a disease. If images obtained by the system were not suitable for disease identification, the system would continue to operate without interruption and would be able to obtain other images of an identified disease for evaluation.

The control subsystem for the rover was checked to make sure it could handle unexpected user input. The web server was able to effectively interpret repeated move commands, abruptly change direction, and stop a rover in a manner that did not disrupt the rover's ability to maintain stability. This helped provide reliable rover navigation for all of the tests.

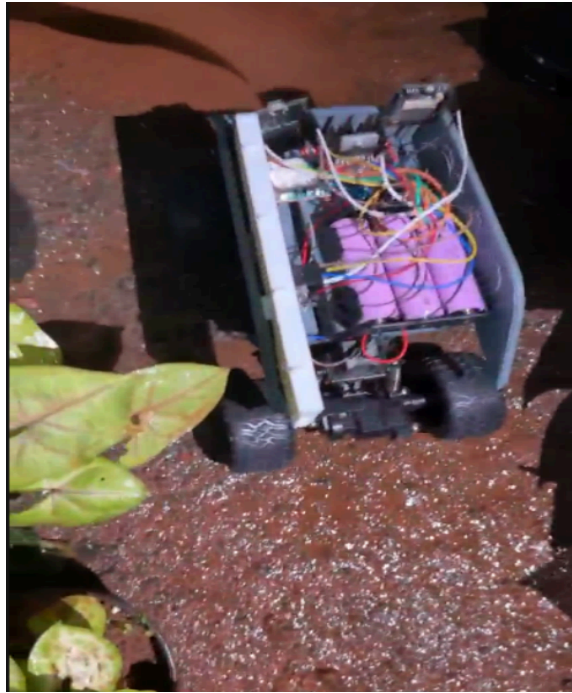


Fig 5.1: System operation under different testing conditions including sensor monitoring, image capture, and rover navigation.

Overall, the implemented system demonstrated reliable performance under various operational scenarios and was capable of handling common challenges encountered in agricultural environments.

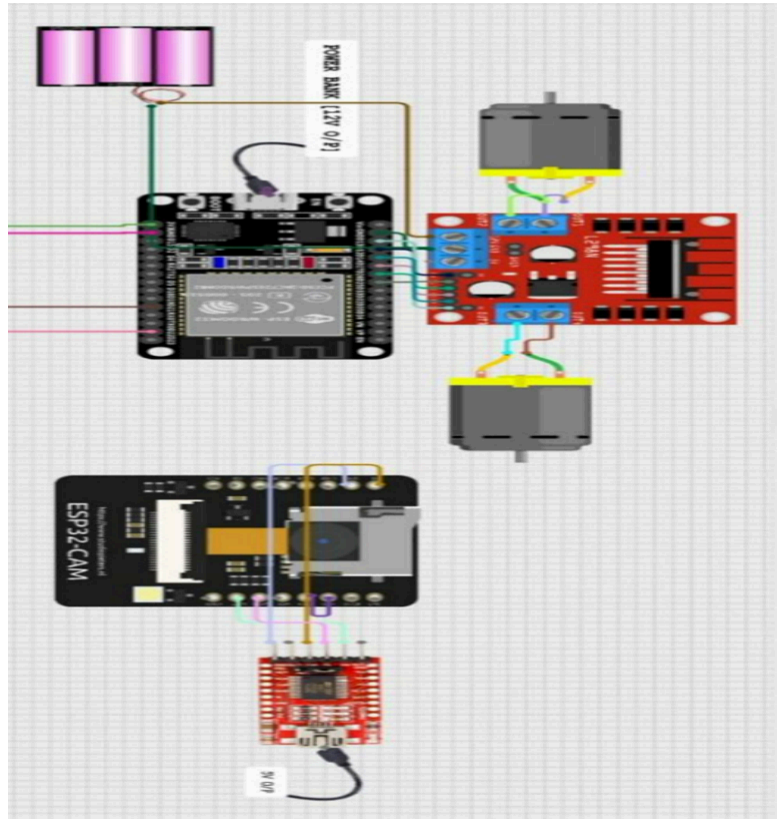


Fig 5.2: Representative connection diagram

5.3 Comparative Assessment

The proposed prototype system integrates environmental monitoring/analysis, soil analysis, remote navigation/l, image-based disease identification, GPS tracking, and intelligent recommendation generation into a single platform. A comparison with conventional agricultural monitoring methods indicates numerous advantages of the proposed system.

Current farming methods depend mostly on manually checking and periodically assessing field crop conditions, which is a time-consuming and labor-intensive process that can lead to late identification of crop diseases and adverse soil conditions. The proposed prototype system solves the timeliness issue by allowing continuous monitoring and easy access to current field information utilizing a web-based dashboard. Current sensor-based monitoring systems typically only gather and monitor environmental factors. While these types of monitoring systems may provide helpful data about soil and weather conditions, they lack the capability to visually assess crop problems or provide any help in determining whether there are crop problems.

The proposed prototype system expands the capabilities of traditional agricultural monitoring systems by integrating the ESP32-CAM and disease classification module, allowing visual crop assessment as part of the monitoring system capabilities. In addition, almost all agricultural monitoring solutions supply raw data only from their sensors. The proposed solution provides added value by using an intelligent recommendation system in conjunction with an intelligent recommendation engine (designed to recommend potential

solutions such as crop management practices and appropriate fertilizers to use) to allow for faster and more accurate decision-making for crop health issues, thereby reducing the need for consulting with experts regarding cropping conditions. In addition, the use of GPS tracking capabilities allows for the ability to geographically reference the locations of the diseases and the observations made in the crops to help facilitate future growth of precision agriculture, which involves targeting actions to specific points in a field instead of the entire area of the field. There is evidence from field trials showing that an improved way of monitoring and accessing agricultural practices and making decisions may replace outdated methods. Through the convergence of the Internet of Things (IoT), computer vision and intelligent recommendation systems, these technologies are developing a viable and scalable method for enabling and supporting smart farms.

APPENDIX

APPENDIX A: Rover Control Code

```
#include <WiFi.h>

#include <WebServer.h>

#include <TinyGPS++.h>

const char* ssid = "Samarjeet23";

const char* password = "samyeet123";

WebServer server(80);

// GPS

TinyGPSPlus gps;

HardwareSerial GPSSerial(2);

// L298N Pins

#define ENA 15

#define IN1 12

#define IN2 18

#define ENB 26

#define IN3 14

#define IN4 27

void setSpeed() {

    analogWrite(ENA, 255);

    analogWrite(ENB, 255);
```

```

}

void stopMotors() {

    digitalWrite(IN1, LOW);

    digitalWrite(IN2, LOW);

    digitalWrite(IN3, LOW);

    digitalWrite(IN4, LOW);

}

void moveForward() {

    digitalWrite(IN3, LOW);

    digitalWrite(IN4, LOW);

    digitalWrite(IN1, HIGH);

    digitalWrite(IN2, LOW);

}

void moveBackward() {

    digitalWrite(IN1, LOW);

    digitalWrite(IN2, HIGH);

    digitalWrite(IN3, HIGH);

    digitalWrite(IN4, HIGH);

}

void turnLeft() {

    digitalWrite(IN1, HIGH);

```

```

digitalWrite(IN2, LOW);

digitalWrite(IN3, HIGH);

digitalWrite(IN4, LOW);

void turnRight() {

    digitalWrite(IN1, LOW);

    digitalWrite(IN2, HIGH);

    digitalWrite(IN3, LOW);

    digitalWrite(IN4, HIGH);

}

void handleGPS() {

    String json = "{}";

    if (gps.location.isValid()) {

        json += "\"latitude\":" + String(gps.location.lat(), 6) + ",";

        json += "\"longitude\":" + String(gps.location.lng(), 6) + ",";

        json += "\"satellites\":" + String(gps.satellites.value());

    } else

        json += "\"latitude\":0,";

        json += "\"longitude\":0,";

        json += "\"satellites\":0";

    }

    json += "}";

```



```
server.send(200, "application/json", json);

}

void handleRoot() {

    String page = R"rawliteral(

<!DOCTYPE html>

<html>

<head>

<meta name="viewport" content="width=device-width, initial-scale=1">

<style>

body{

font-family:Arial;

text-align:center;

margin-top:40px;

}

button{

width:120px;

height:60px;

font-size:20px;

margin:10px;

}

</style>
```

```
</head>

<body>

<h1>AgroNet Rover</h1>

<p>

<a href="/forward"><button>Forward</button></a>

</p>

<p>

<a href="/left"><button>Left</button></a>

<a href="/stop"><button>Stop</button></a>

<a href="/right"><button>Right</button></a>

</p>

<p>

<a href="/backward"><button>Backward</button></a>

</p>

<p>

<a href="/gps"><button>GPS Data</button></a>

</p>

</body>

</html>

)rawliteral";
```

```
server.send(200, "text/html", page);
```

```

}

void setup() {

  Serial.begin(115200);

  GPSSerial.begin(9600, SERIAL_8N1, 16, 17);

  pinMode(ENA, OUTPUT);

  pinMode(IN1, OUTPUT);

  pinMode(IN2, OUTPUT);

  pinMode(ENB, OUTPUT);

  pinMode(IN3, OUTPUT);

  pinMode(IN4, OUTPUT);

  setSpeed();

  stopMotors();

  WiFi.begin(ssid, password);

  Serial.print("Connecting");

  while (WiFi.status() != WL_CONNECTED) {

    delay(500);

    Serial.print(".");

  }

  Serial.println();

  Serial.println("WiFi Connected");

  Serial.print("IP Address: ");

```

```
Serial.println(WiFi.localIP());

server.on("/", handleRoot);

server.on("/gps", handleGPS);

server.on("/forward", []() {

    moveForward();

    server.sendHeader("Location", "/");

    server.send(303);

});

server.on("/backward", []() {

    moveBackward();

    server.sendHeader("Location", "/");

    server.send(303);

});

server.on("/left", []() {

    turnLeft();

    server.sendHeader("Location", "/");

    server.send(303);

});

server.on("/right", []() {

    turnRight();

    server.sendHeader("Location", "/");
```

```

    server.send(303);

});

server.on("/stop", []) {

    stopMotors();

    server.sendHeader("Location", "/");

    server.send(303);

});

server.begin();

Serial.println("Server Started");

}

void loop() {

    while (GPSSerial.available()) {

        gps.encode(GPSSerial.read());

    }

    if (gps.location.isUpdated()) {

        Serial.print("Latitude: ");

        Serial.println(gps.location.lat(), 6);

        Serial.print("Longitude: ");

        Serial.println(gps.location.lng(), 6);

        Serial.print("Satellites: ");

        Serial.println(gps.satellites.value());

```

```

    Serial.println("-----");

}

server.handleClient();

}

```

APPENDIX B: ACRONYMS

Acronym	Full Form
ADC	Analog to Digital Converter
API	Application Programming Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DHT11	Temperature and Humidity Sensor
ESP32	Espressif 32-bit Microcontroller
GPS	Global Positioning System
Wi-Fi	Wireless Fidelity
SMU	Static Monitoring Unit
URL	Uniform Resource Locator
USB	Universal Serial Bus
I2C	Inter Integrated Circuit
JSON	JavaScript Object Notation
ML	Machine Learning
RTC	Real Time Clock
WWW	World Wide Web
XML	Extensible Markup Language

APPENDIX C: Hardware Specifications

ESP32 Development Board

- Dual-core 32-bit Xtensa LX6 Microcontroller
- Operating Voltage: 3.3V
- Clock Frequency: Up to 240 MHz
- Built-in Wi-Fi (802.11 b/g/n) and Bluetooth (BLE)

ESP32-CAM Module

- ESP32-based Camera Development Board
- OV2640 2MP Camera Sensor
- Supports Real-Time Video Streaming
- Built-in Wi-Fi and Bluetooth Connectivity

Soil Moisture Sensor

- Measures Water Content in Soil
- Analog and Digital Output Support
- Operating Voltage: 3.3V – 5V
- Fast Response Time

L298N Motor Driver Module

- Dual H-Bridge Motor Driver
- Operating Voltage: 5V – 35V
- Controls Two DC Motors Simultaneously
- Supports Forward and Reverse Motor Operation

Lithium-Ion Battery Pack

- Rechargeable Power Source for Rover Unit
- Output Voltage: 7.4V – 12V (Configuration Dependent)
- High Energy Density
- Lightweight and Portable Design

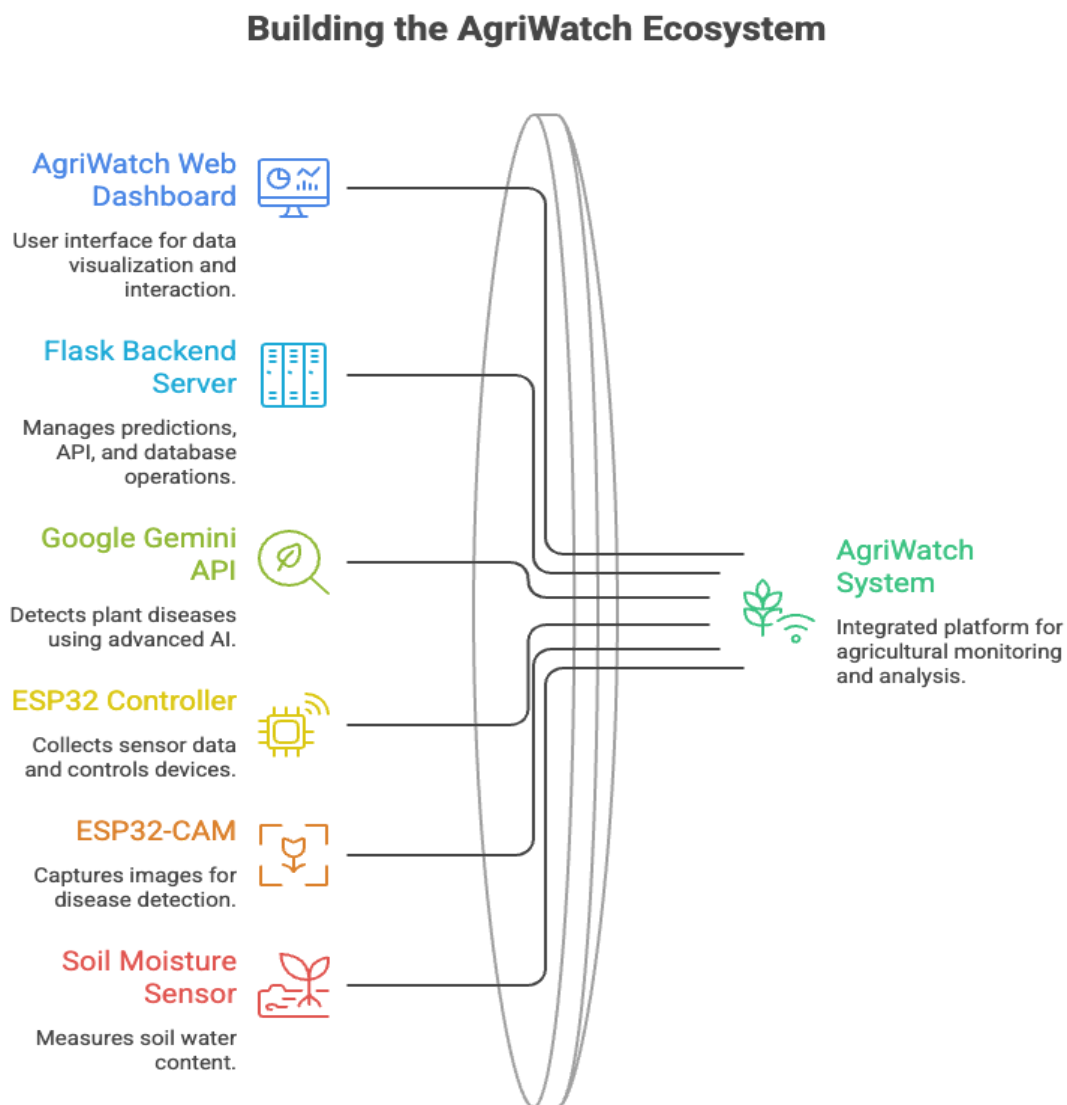
Jumper Wires and Breadboard

- Used for Circuit Prototyping and Connections
- Reusable and Easy to Configure
- Supports Rapid Hardware Development
- Compatible with ESP32 and Sensor Modules

AgriWatch – AI-Powered Plant Health Monitoring System

AgriWatch is an AI-powered smart agriculture platform that combines IoT sensing, autonomous image capture, and deep learning-based disease detection to help farmers monitor crop health efficiently. The system utilizes an ESP32-CAM mounted on a rover along with environmental sensors to collect real-time field data. Captured plant images are analyzed using a TensorFlow Lite model, and sensor readings are displayed through an interactive web dashboard to provide disease diagnosis, environmental monitoring, and crop management recommendations.

Architecture:



Made with Napkin

The system comprises three core components:

1. ESP32 IoT Device

Interfaces with all field sensors and the ESP32-CAM. Collects environmental parameters including soil moisture, temperature, humidity, and pH values while capturing plant images for disease analysis.

2. Flask Backend

Handles image uploads, processes sensor data, communicates with the web interface, and generates disease diagnosis along with treatment recommendations.

3. Web Dashboard

Built using HTML, CSS, and JavaScript. Displays uploaded plant images, disease prediction results, confidence scores, sensor values, and recommended remedies through an intuitive interface.

Key Features

- **AI Disease Detection:** Uses a Gemini API Lite deep learning model to identify plant diseases from uploaded leaf images.
- **Real-time Sensor Monitoring:** Continuously monitors soil moisture, temperature, humidity, and pH values.
- **ESP32-CAM Image Capture:** Captures plant leaf images directly from the field for automated disease analysis.
- **Treatment Recommendation:** Displays appropriate remedies and preventive measures based on the detected disease.
- **Responsive Web Dashboard:** Provides an easy-to-use dashboard for viewing predictions and environmental conditions.
- **Lightweight Edge AI:** Uses TensorFlow Lite for faster inference with lower computational requirements.
- **Environmental Monitoring:** Helps farmers understand crop conditions using multiple IoT sensors.
- **Scalable Architecture:** Designed for future integration with cloud storage, weather APIs, and mobile applications.

Required Arduino Libraries

Install the following libraries before uploading the ESP32 firmware.

1. WiFi
2. ESP32 Camera
3. DHT Sensor Library
4. Adafruit Unified Sensor
5. ArduinoJson

Setup & Execution Guide

1. Backend Installation

Navigate to the project folder

```
cd AgriWatch
```

Install dependencies

```
pip install -r requirements.txt
```

Run the Flask application

```
python app.py
```

The server starts on

```
http://127.0.0.1:5000
```

2. ESP32 Configuration & Upload

1. Open the Arduino sketch.
2. Enter your Wi-Fi credentials.

```
const char ssid = "YOUR_WIFI";
```

```
const char password = "YOUR_PASSWORD";
```

3. Select

Tools → Board → ESP32 Dev Module

or

AI Thinker ESP32-CAM

depending on your hardware.

4. Select the correct COM port.
5. Click **Upload**.
6. Open the Serial Monitor at **115200 baud** to verify sensor readings and Wi-Fi connection.



Dashboard Operations & Walkthrough

The AgriWatch dashboard provides a complete overview of plant health and environmental conditions.



Disease Prediction

Upload or capture a plant leaf image to receive:

- Disease name
 - Prediction confidence
 - Disease description
 - Treatment recommendation
-



Environmental Monitoring

Displays:

- Temperature
 - Humidity
 - Soil Moisture
 - Soil pH
-



Live Image Capture

The ESP32-CAM captures real-time images of crops for continuous monitoring and disease diagnosis.



Treatment Recommendation

Based on the AI prediction, the dashboard suggests:

- Recommended pesticides/fungicides
 - Preventive measures
 - Crop management practices
-

Future Cloud Analytics

The architecture supports future integration with:

- Weather API
- Cloud database
- Historical crop analytics
- Mobile application
- Smart irrigation automation

